

Error anatomy of a frozen-trunk $\Delta\Delta G$ predictor

Which mutation classes are hard, on held-out data and on two blind corpora — and where the learned representation actually beats a substitution matrix.

Aggregate accuracy hides a structured error. Held-out performance ($\rho = 0.784$, MAE = 0.44 kcal/mol over 12,359 mutations) is not uniform across the substitution space, but most of the apparent structure is an effect-size artifact: classes containing larger $\Delta\Delta G$ values show larger absolute errors and identical *relative* accuracy. Three things survive normalisation. (i) The dominant deficit is the **stabilizing tail**: error is 2.1× the class's own spread and the bias is 0.56 kcal/mol — stabilizing mutations are called destabilizing. (ii) On blind transfer, **proline in either direction and mutations leaving glycine** are genuinely harder, while **mutations to glycine are not**. (iii) The representation's advantage over a plain amino-acid lookup is **not uniform**: it is largest for core-packing substitutions and near zero when the mutation preserves the chemistry.

1. Question

Holdout designs in this series measure *generalization to unseen classes* — unseen substitution types, source residues, target residues. That is a different question from *which mutations are hard*, which had not been asked on a blind set. The literature on $\Delta\Delta G$ prediction makes two specific predictions worth testing: larger errors for glycine and proline substitutions, and degraded accuracy at buried positions. A third question only becomes askable once a matched trivial baseline exists: for which mutations does a learned structural representation contribute anything beyond knowing which residue replaced which?

2. Method

Predictions come from a regressor on frozen structure-model embeddings. Three evaluation sets are used: **12,359 out-of-fold predictions** from 5-fold grouped cross-validation on the training corpus (grouping by protein, so no protein is ever in both folds), and **two blind corpora** never seen in training — S669 (541 variants over 62 proteins) and a homology-filtered FireProt subset (3102 variants over 130 proteins). The blind corpora are scored with the readout selected for transfer in the companion feature study (the pair-track diagonal at the mutated site); the out-of-fold set uses the in-distribution readout.

Four conventions make the class comparisons interpretable:

- **Protein-centred error.** Every table is computed on the raw error and on the error with each protein's mean error removed, so a class effect cannot be an artifact of that class sitting in badly-calibrated proteins.
- **Normalisation by class spread.** Class MAE is read against that class's own standard deviation of true $\Delta\Delta G$. Without it, any class containing larger effects looks harder by arithmetic alone.
- **Burial from the model's own predicted distogram** — expected number of residues within 10 Å of the mutated site, excluding trivial backbone neighbours. No external structures or secondary-structure assignment are required.
- **Cluster bootstrap over proteins** for every confidence interval, since variants within a protein are not independent.

3. Burial does not degrade the model

Burial tertile	n	MAE	ρ	sd(true)	MAE ÷ sd
buried	3,730	0.60	0.79	1.23	0.49
mid	3,729	0.42	0.74	0.87	0.48
exposed	3,730	0.35	0.69	0.73	0.48

Buried sites carry 0.60 kcal/mol of error against 0.35 at exposed sites — a 1.7× gap that would ordinarily be read as "the model is worse in the core". It is not: relative accuracy is flat (0.49 / 0.48 / 0.48), and *ranking* is in fact best at buried sites (ρ 0.79 against 0.69). Burial scales the errors because it scales the effects; it does not degrade the model. The single interaction cell that does stand out is **buried glycines** (MAE ÷ sd 0.64, $n = 81$, against 0.48 for buried non-glycines) — the same regime in which a physics-based method independently breaks down.

4. The dominant deficit is the stabilizing tail

Effect direction	n	MAE	bias	ρ	sd(true)	MAE ÷ sd
stabilizing	535	0.64	0.56	0.27	0.30	2.15
destabilizing	6,120	0.54	-0.21	0.71	0.89	0.61
neutral	5,704	0.32	0.13	0.36	0.25	1.29

This is the one class effect that is not an arithmetic artifact, and it runs the wrong way for protein engineering. The model's error on stabilizing mutations is roughly twice the spread of the class itself, the bias is strongly positive — stabilizing variants are predicted less stabilizing than they are — and within-class ranking nearly collapses (ρ 0.27 against 0.71 for destabilizing variants). Amplitude compression toward the mean, visible directly in the predicted-versus-measured panel below, is the mechanism.

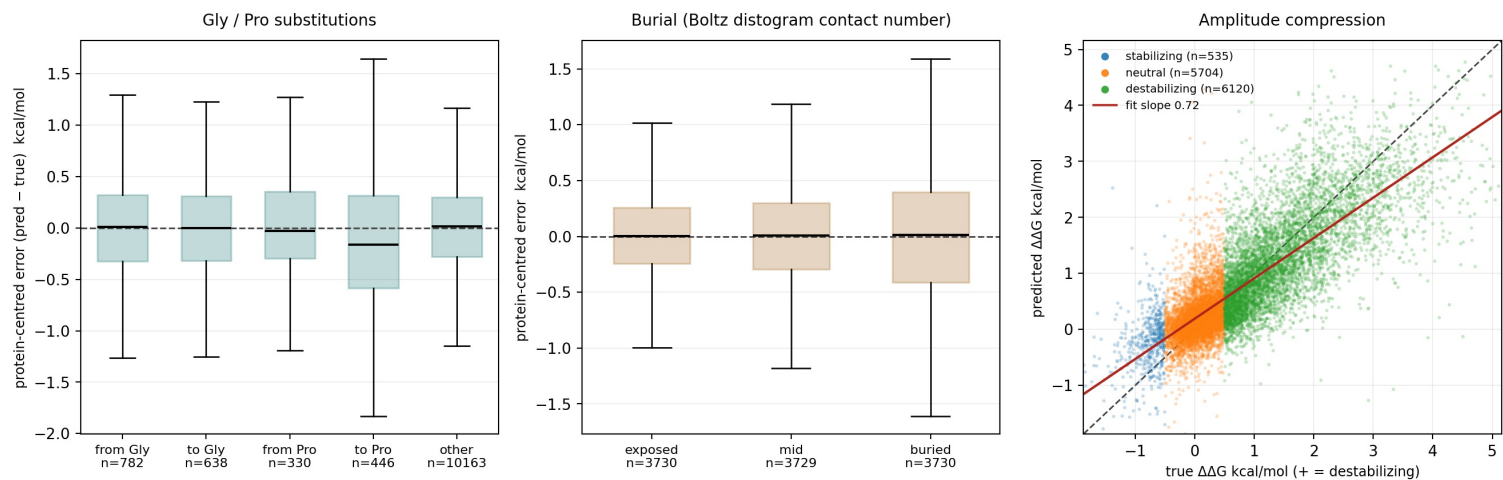


Figure 1. Held-out predictions, 12,359 mutations. Left: protein-centred error by glycine/proline class. Middle: by burial tertile. Right: predicted against measured $\Delta\Delta G$ coloured by effect direction, with the amplitude-compression fit — the slope below unity is what produces the stabilizing-tail bias.

5. Which classes stay hard on blind transfer

Classes are contrasted against all other mutations in the same corpus, as a difference in $MAE \div sd$, with a cluster bootstrap over proteins. Both blind corpora are reported.

Class	n	FireProt $\Delta(MAE \div sd)$	n	S669 $\Delta(MAE \div sd)$
→ Pro	80	+0.099 [+0.049, +0.740]	6	-0.012 [-0.042, +0.933]
from Pro	82	+0.098 [+0.010, +0.216]	6	+0.248 [+0.171, +0.537]
from Gly	157	+0.085 [+0.007, +0.245]	12	+0.025 [-0.240, +1.163]
→ Gly	274	-0.058 [-0.116, +0.013]	55	-0.001 [-0.141, +0.172]
from aromatic (F/W/Y)	352	+0.016 [-0.043, +0.104]	64	+0.119 [+0.001, +0.299]
X → Ala	723	+0.004 [-0.042, +0.045]	220	-0.101 [-0.183, -0.007]
near-isosteric ($ \Delta Vol < 30 \text{ \AA}^3$)	1529	+0.070 [+0.028, +0.113]	252	+0.065 [-0.007, +0.138]

Bold = the 95 % interval excludes zero. Positive = the class is harder than the rest.

The proline and glycine result reproduces the ordering seen in-distribution (→Pro 0.56, from-Gly 0.51, from-Pro 0.50 against 0.43 for everything else), and sharpens it: **the weak class is leaving glycine, not arriving at it** — →Gly is, if anything, easier than average (0.46 in-distribution, and negative on both blind corpora).

Near-isosteric substitutions are hard, and not only by the normalisation. They also degrade in *ranking*, which has no such floor: ρ 0.589 against 0.709 for the rest on FireProt, and 0.438 against 0.638 on S669. This is the transfer-side face of the same phenomenon as the stabilizing tail: the model resolves large effects and blurs small ones.

Class-level questions need protein-level replication, not variant counts. The two blind corpora agree on the sign of the near-isosteric and →Gly results, but only *from Pro* clears zero on both. S669 is underpowered for most cells (→Pro $n = 6$, from Gly $n = 12$) and disagrees on the two classes where it does have sample size. The per-residue error ranking is the clearest demonstration: between the two corpora its rank correlation is +0.05 ($p = 0.84$, $k = 17$) — that is, a "hardest source residues" list read off 541 variants over 62 proteins carries no information about the next corpus. What does replicate is within-class ranking quality (+0.67 ($p = 0.05$, $k = 9$)). **Per-residue error rankings should not be quoted from a single benchmark of this size.**

6. Where the representation beats a substitution matrix

All of the above measures where error is large. A different and more actionable question is where the learned representation contributes anything at all. Scoring the same variants with a 40-dimensional one-hot encoding of the substitution identity — a model that can only learn an average effect per residue pair, with no structure and no context — gives a matched trivial baseline, and

$$\text{skill} = 1 - MAE(\text{representation}) \div MAE(\text{one-hot substitution})$$

measures what the embedding adds. Pooled, skill is **+0.224** on FireProt and +0.145 on S669 — the representation is not an amino-acid lookup. But it is not uniformly better than one either:

Source residue	skill, S669	skill, FireProt	Substitution	n (FireProt)	skill, FireProt
Gln	-0.289	-0.029	Tyr → Phe	64	-0.031
Trp	0.019	0.040	Trp → Ala	14	-0.014
Thr	0.007	0.146	Trp → Phe	24	0.018
Lys	0.175	0.127	Lys → Arg	15	0.028
Ile	0.271	0.179	Val → Ala	75	0.217
Leu	0.252	0.281	Ile → Ala	30	0.201
Val	0.284	0.266	Leu → Ala	58	0.345

Two residues sit at zero on *both* blind corpora: mutations away from **glutamine** and from **tryptophan**, where the representation matches but does

not beat the substitution matrix. At the other end, mutations away from the aliphatics — Val, Leu, Ile — and from Ala carry most of the model's advantage. By substitution pair the split is cleaner still: chemistry-preserving replacements (Tyr→Phe, Trp→Phe, Lys→Arg) sit at zero skill, while large-hydrophobic-to-alanine truncations, the classic core-packing perturbation, are where the representation is worth the most.

The full ordering of skill across residues replicates only weakly between the two corpora (+0.40 (p = 0.11, k = 17)), so the extremes are the defensible claim, not the rank order.

Error anatomy on transfer, with the transfer-best readout (pair-track diagonal, results/14)

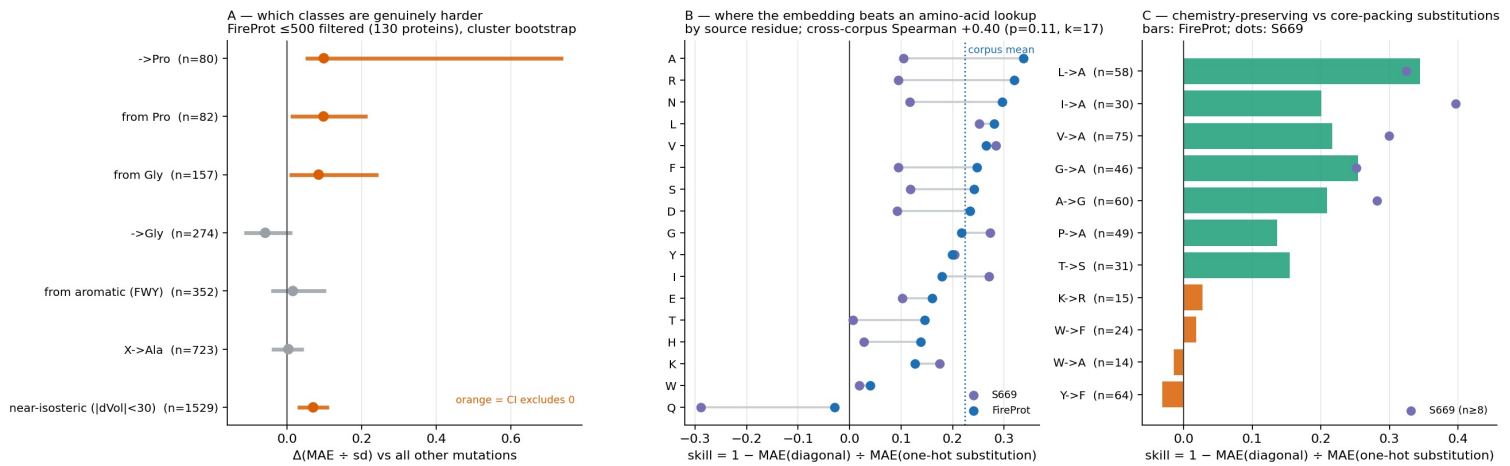


Figure 2. Blind transfer, both corpora. **A:** class contrasts in MAE + sd against all other mutations, with cluster-bootstrap intervals over the 130 FireProt proteins. **B:** skill over the one-hot substitution baseline by source residue, on both corpora. **C:** the same by substitution pair, contrasting chemistry-preserving replacements against core-packing truncations.

7. Interpretation

Read together, the three surviving effects are one effect seen from different angles. The stabilizing tail, the near-isosteric substitutions and the chemistry-preserving pairs are all **small-amplitude** perturbations, and the model compresses amplitudes: it separates large destabilization from the rest confidently and blurs everything inside a narrow band. Burial, which the literature flags, turns out to be a proxy for amplitude rather than a difficulty axis of its own — once amplitude is controlled, it disappears entirely.

The skill decomposition adds a constraint on *why*. Where the mutation changes residue type dramatically — a large hydrophobic truncated to alanine — a substitution-identity average is already informative and the structural representation improves substantially on it. Where the mutation preserves chemistry and the answer depends on packing detail at that specific site, the representation falls back to roughly what the substitution identity alone would predict. That is the signature of a readout carrying residue-type information plus coarse context, rather than a fine-grained model of the local environment. It also says the remaining headroom is not in the loss or in the regressor: a reweighting that emphasises the stabilizing tail cannot manufacture discrimination that the frozen features do not encode.

Methodologically, two of the conventions used here changed conclusions rather than decorating them. Normalising class error by class spread removed the burial effect entirely and reversed the reading of every raw-MAE table. Requiring class-level claims to replicate across two blind corpora, with protein-level rather than variant-level resampling, removed the per-residue error ranking — the kind of result that is easy to report from a single benchmark and that carries no information about the next one.